Explainable Exoplanet Triage for Noisy TESS Light Curves

1. Title

I am building an AI-based pipeline that would detect exoplanet-like transit signals in noisy TESS light curves, reject obvious false positives early, and classify the remaining candidates into planets, eclipsing binaries, blends, or other astrophysical signals.
Our goal would not be just detection. We would build a practical vetting system that would be scalable, interpretable, and useful for real survey data.

Referred to this research paper it have really good insights academic.oup

2. The problem

TESS light curves are noisy, crowded, and difficult to interpret. A real transit can be extremely small, and many non-planetary signals can look similar to it. That means a simple detection model would not be enough; we would need a system that could detect, reject, classify, and explain.

3. Why existing ideas fall short

Most existing solutions would do one of two things: they would either search for periodic dips, or they would classify already-found candidates. What they often miss is a strong triage layer a stage that would quickly reject flat or clearly non-informative light curves so the model could focus only on meaningful candidates. That is where our idea would stand out.academic.

4. Core idea

I am proposing a three-stage pipeline. First, we would detect periodic dips using a classical method like BLS or Transit Least Squares. Second, we would use an AI model — CNN or Transformer to classify the candidate signal. Third, we would add a confidence-gated rejection step so the model could automatically say that a curve is not a useful candidate when there is no meaningful transit evidence.academic.

5. What makes us different

Our uniqueness would be that we are not just hunting planets we are building an exoplanet triage system. We would design it to reject obvious no-signal curves early, classify false positives more accurately, explain why a curve was accepted or rejected, and estimate transit parameters only when the signal is truly meaningful. This would be different from a generic classifier because it would behave more like a scientific assistant than a black box.academic.

6. How we solve the problem

We would first clean and normalize the TESS light curves, then search for periodic dips using BLS or Transit Least Squares. After that, we would pass only candidate curves into the AI classifier. The classifier would identify transits, eclipsing binaries, blends, or noise-like signals. For high-confidence transits, we would fit the curve to estimate the period, depth, and duration, and we would also provide a significance score.transitleastsquares.

7. USP

My unique selling point would be confidence-gated, explainable rejection. Instead of only telling the user what the signal is, we would also tell them when a curve should be rejected early because it does not contain a meaningful transit pattern. This would help reduce false alarms, save computation, and make the system much more trustworthy for large astronomical datasets.academic.

8. Technologies we would use

We would build this using Python, Lightkurve, Astropy, NumPy, Pandas, SciPy, scikit-learn, and either TensorFlow or PyTorch. For detection, we would use BLS or Transit Least Squares. For classification, we would use a CNN as the baseline and optionally compare it with a Transformer for better interpretability. For visualization, we would use Plotly or Matplotlib to show original, detrended, and phase-folded light curves.academic.

9. Workflow

Our workflow would be simple and practical. We would download TESS light curves, preprocess and normalize the data, detect candidate periods, reject empty or flat curves, classify the remaining signals, fit transit parameters, and display the result with confidence and explanation. This workflow would be built for hackathon execution, but it would also be scalable enough to look like a real mission-support tool.transitleastsquares.

10. Expected impact

Our solution would help reduce the time needed to vet large numbers of TESS light curves. It would also reduce false positives, improve candidate prioritization, and make the final output easier to trust because it would include both prediction and explanation. In the long term, we believe this approach could support faster and smarter exoplanet discovery pipelines.

11. Closing

We are not just building a planet detector. We are building a practical, explainable, and scalable triage system for noisy astronomical data. That is why our project would be different, useful, and strong enough to stand out in a large competition like the ISRO Bharatiya Antariksh Hackathon

What matters most

For this dataset, the hardest part is not just finding dips — it is **telling real planetary transits apart from everything that imitates them**. That is exactly where many teams fail, because a model can look good on easy cases but still confuse planets with false positives.academic.

Why this is the core challenge

The data is also **imbalanced**, meaning there are far fewer true exoplanet cases than non-planet cases. That means a model can become biased toward predicting “no planet” too often, or it can overfit to the few positive examples and fail on new light curves. Another issue is that some planets only show one or two transits, especially for longer periods, which makes detection and validation much harder